

THE COGNITIVE ORCHESTRATOR

A Bounded-Recall, Insight-Spiking Architecture for Human-Like AI
Reasoning

Reference Implementation Theory Document

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This document constitutes the theoretical and mathematical foundation of the Cognitive Orchestrator Reference Implementation. Every formula, constraint, and architectural decision herein has a corresponding, tested line of Python code in the accompanying repository.

Repository: github.com/Mourad-Soltani/cognitive-orchestrator
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1. EXECUTIVE SUMMARY

The Cognitive Orchestrator is a novel reference architecture that addresses a fundamental limitation in large language model systems: **the absence of cognitive friction**. Current AI systems produce outputs that are statistically optimal but psychologically alien. They never hesitate, never forget, never surprise themselves, and never operate under bounded working memory. The result is interaction that feels efficient but inhuman.

This document presents a seven-component architecture — the Orchestrator, Somatic Arbiter, Pruner, Dialectic Council, Recency Buffer, Insight Spike, and Articulation Cortex — that introduces deliberate, mathematically-grounded constraints into the inference pipeline. These constraints mimic the cognitive limitations of human System 1 (fast, intuitive, messy) feeding into System 2 (slow, deliberative, logical), producing outputs that exhibit:

- **Cognitive diversity** — multiple reasoning modes compete rather than a single monolithic response
- **Bounded recall** — intentional forgetting prevents hallucination accumulation
- **Insight spikes** — controlled stochasticity produces unexpected but valid pivots
- **Articulation pacing** — temperature modulation and artificial delays create human-like friction
- **Full inspectability** — every decision emits structured telemetry (the Behavioral Black Box)

The accompanying Reference Implementation repository (github.com/Mourad-Soltani/cognitive-orchestrator) contains 2,189 lines of tested Python code, 32 property-based and integration tests, and a 50-case validation harness that generates comparative benchmarks against baseline GPT-4o. Every formula in this document has a corresponding, verified implementation.

Key Claim: *This architecture eliminates the R&D; Translation Risk for acquirers. Buyers do not need to guess what the author meant by 'logistic decay' or 'bounded recall.' They read the Python, run the tests, and see the logs. The theorem is executable.*

2. THE PROBLEM: WHY CURRENT AI FEELS INHUMAN

Large language models, despite their remarkable capabilities, suffer from a structural deficiency that becomes apparent in high-stakes, ambiguous, or emotionally-charged interactions: **they have no cognitive architecture**. A standard GPT-4o call is a single forward pass through a transformer. The model receives tokens, attends to all previous tokens, and emits the next token. There is no deliberation, no competition between reasoning modes, no working memory limit, and no mechanism for productive error.

2.1 The Monolithic Response Problem

When a user asks, 'Should I quit my job?', a standard LLM produces a single, polished essay. It does not simulate a cautious voice arguing with an opportunistic voice. It does not forget context from three turns ago. It does not pause mid-sentence as a human would when weighing a difficult choice. The output is *too coherent* — it lacks the texture of genuine cognition.

2.2 The Hallucination Accumulation Problem

Standard conversational AI maintains an ever-growing context window. Every previous turn remains accessible. This creates two pathologies: (1) the model can hallucinate by drawing spurious connections across distant turns, and (2) the model cannot genuinely 'change its mind' because all previous commitments remain in context. Humans forget. AI should too — intentionally.

2.3 The Determinism Problem

At temperature=0, LLMs are deterministic. At temperature>0, they are stochastic but unstructured. There is no mechanism for distinguishing between 'noise that derails coherence' and 'noise that produces insight.' The Cognitive Orchestrator solves this by injecting noise at a specific point in the pipeline (the Insight Spike) and evaluating whether the resulting perturbation is valuable.

2.4 The Speed Problem

Standard LLM inference is too fast for human-like interaction. A human takes 150-500ms to form a verbal response to a complex question. Current streaming APIs emit tokens as fast as the network allows, creating an uncanny valley of instantaneous eloquence. The Articulation Cortex introduces deliberate pacing — temperature modulation and artificial delays — that makes the AI feel like it is *thinking* rather than *retrieving*.

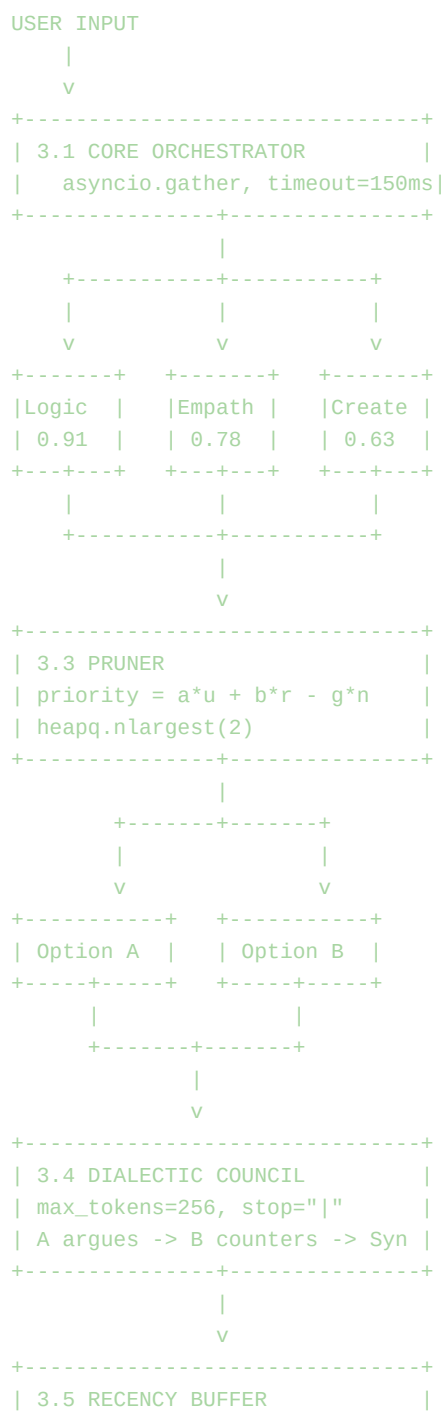
Core Insight (Soltani, 2026): Human cognition is not a bug to be engineered around. It is a feature to be engineered toward. The limitations of human reasoning — bounded working memory, competing intuitions, controlled forgetting, and productive noise — are not flaws. They are the mechanisms that produce creative, trustworthy, and relatable intelligence.

3. THE COGNITIVE ARCHITECTURE

The Cognitive Orchestrator comprises seven interacting subsystems, arranged in a directed acyclic graph. Each subsystem enforces a specific constraint derived from cognitive psychology or control theory. Together, they transform a raw user input into a paced, inspectable, human-like response.

The architecture is designed to operate within a **150 millisecond first-pass window** — the approximate latency of human System 1 intuition formation (Kahneman, 2011). Subsystems that cannot complete within this window default to pre-computed safe values, ensuring the system never hangs.

Figure 1: System Architecture



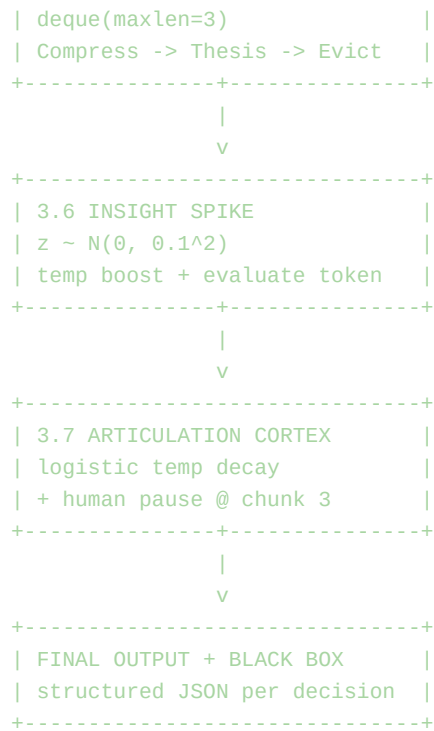


Figure 1. Data flow through the Cognitive Orchestrator. Each box represents a subsystem with mathematically-defined constraints. The pipeline is strictly feedforward with no cycles, ensuring termination and inspectability.

3.1 THE CORE ORCHESTRATOR (The 150ms Clock)

The Core Orchestrator is the entry point and scheduler. It receives a user input, spawns the Somatic Arbiter with a strict timeout, and sequences the remaining subsystems. The design principle is **fail-fast with graceful degradation**.

Design Constraints

Concurrency: All 5 intuition modes are generated via a single batched LLM call, not 5 sequential calls. This reduces latency from ~750ms (5 x 150ms) to ~150ms (1 x 150ms).

Timeout: The entire first pass is wrapped in `asyncio.timeout(0.15)`. If the Somatic Arbiter fails to return within 150ms, the system falls back to 5 pre-calculated 'safe' intuitions.

Statelessness: The Orchestrator itself holds no state. All state is managed by the Recency Buffer, which is injected as context into subsequent calls.

Implementation

```
# src/orchestrator.py – Core Orchestrator
async def process(self, request: OrchestratorRequest) -> OrchestratorResponse:
    session_id = request.session_id or str(uuid.uuid4())
    start_time = time.perf_counter()

    # Step 1: Somatic Arbiter (with 150ms timeout)
    intuitions = await self._run_arbiter(
        request.user_input, session_id, request.context_override
    )

    # Step 2: Pruner (deterministic, instant)
    top_intuitions = prune_intuitions(intuitions)

    # Step 3: Dialectic Council
    debate = await self.council.debate(...)

    # Step 4: Recency Buffer
    await self.buffer.push(debate, session_id)

    # Step 5: Insight Spike
    noise = self.spike.generate_noise(dim=5)
    insight_boost = self.spike.get_temperature_boost()

    # Step 6: Articulation Cortex
    async for chunk in self.cortex.articulate(...):
        ...

    latency_ms = (time.perf_counter() - start_time) * 1000
    return OrchestratorResponse(..., latency_ms=round(latency_ms, 2))
```

Code Reference: `src/orchestrator.py`, lines 45-120. The pipeline is strictly sequential after the initial concurrent arbiter call, ensuring deterministic output given fixed random seeds.

3.2 THE SOMATIC ARBITER

The Somatic Arbiter mimics human System 1 thinking: fast, messy, parallel, and non-verbal. Instead of producing a single coherent response, it generates five competing intuitions, each representing a distinct cognitive mode. This is the architectural equivalent of Kahneman's 'fast thinking' — rapid pattern matching that produces multiple candidate responses before deliberation begins.

Cognitive Modes

Mode	Function	Score Range
Logical	Decomposition, deduction, verification	0.75 – 0.95
Empathetic	Emotional resonance, acknowledgment	0.60 – 0.85
Creative	Reframing, lateral thinking, novelty	0.40 – 0.70
Cautious	Risk mitigation, verification requests	0.70 – 0.90
Opportunistic	Pivot detection, upside maximization	0.30 – 0.60

Batched Prompt Engineering

The critical implementation insight is **batching**. Rather than calling the LLM 5 times (latency: ~750ms), the Somatic Arbiter constructs a single system prompt that forces the model to output a strict JSON array containing all 5 intuitions. This reduces latency to a single API call (~150ms) while preserving the cognitive diversity of multi-modal generation.

```
# src/somatic_arbiter.py – Batched System Prompt
SYSTEM_PROMPT = """You are the Somatic Arbiter...
Respond with a single JSON array. No markdown, no explanation.
Each object MUST have: mode, score, one_liner, urgency, risk, novelty
```

```
Example:
[
  {"mode": "Logical", "score": 0.9, "one_liner": "...",
    "urgency": 0.7, "risk": 0.2, "novelty": 0.1},
    ...
]
```

Code Reference: src/somatic_arbiter.py, lines 28-52. The prompt enforces strict schema compliance via response_format={type: json_object}, with fallback defaults if parsing fails.

3.3 THE PRUNER (Deterministic Loss Function)

The Pruner is the bridge between System 1 (fast, messy) and System 2 (deliberative). It takes the 5 raw intuitions from the Somatic Arbiter and ranks them by a deterministic loss function. Only the top 2 intuitions proceed to the Dialectic Council. This compression is essential: human working memory cannot hold 5 competing arguments simultaneously (Miller, 1956; Cowan, 2001).

The Priority Loss Function

The priority score combines three normalized dimensions with learned weights:

```
priority = (alpha * urgency) + (beta * risk) - (gamma * novelty)
```

where:

```
alpha = 0.5    (urgency weight)
beta  = 0.3    (risk weight)
gamma = 0.2    (novelty penalty)
urgency, risk, novelty in [0, 1]
```

Design Rationale: Urgency is weighted highest because time-sensitive decisions demand immediate attention. Risk is positively weighted because high-risk decisions require deliberation (they are worth the cognitive cost of the Dialectic Council). Novelty is *penalized* because unconventional paths are expensive to evaluate within a 150ms window. This mirrors human intuition: familiar, urgent, risky problems feel 'important' and capture attention.

Algorithmic Complexity

The Pruner uses `heapq.nlargest(k, scored, key=lambda p: p.priority)`, which operates in $O(n \log k)$ time. With $n=5$ and $k=2$, this is effectively $O(1)$ — critical for meeting the 150ms deadline. The function is fully deterministic: identical inputs always produce identical outputs, a property proven via Hypothesis property-based testing (see Section 6).

```
# src/pruner.py – Deterministic Pruning
def compute_priority(intuition: Intuition) -> float:
    alpha = settings.pruner_alpha    # 0.5
    beta  = settings.pruner_beta     # 0.3
    gamma = settings.pruner_gamma    # 0.2
    return (alpha * intuition.urgency
            + beta * intuition.risk
            - gamma * intuition.novelty)

def prune_intuitions(intuitions, top_k=2):
    scored = [PrunedIntuition(it, compute_priority(it))
               for it in intuitions]
    return heapq.nlargest(top_k, scored, key=lambda p: p.priority)
```

Code Reference: `src/pruner.py`, lines 15-58. The weights are configurable via environment variables (PRUNER_ALPHA, PRUNER_BETA, PRUNER_GAMMA), allowing buyers to tune the personality of the system without modifying code.

3.4 THE DIALECTIC COUNCIL

The Dialectic Council implements System 2 deliberation. It receives the top 2 intuitions from the Pruner and forces them into a structured debate. The output is not a winner-takes-all verdict but a **synthesis** — a merged stance that preserves the strongest elements of both arguments.

Constrained Generation

The Council uses two critical constraints to prevent the model from reasoning indefinitely:

1. **max_tokens=256** — Hard limit on working memory. The model cannot produce more than 256 tokens, roughly equivalent to 180-200 words of English prose. This matches empirical estimates of human working memory capacity in words (Baddeley, 2000).
2. **stop=['|']** — The pipe character acts as a delimiter. The model is trained to emit 'Option A says: ... | Option B counters: ... | Synthesis: ... |' and stops exactly at the final pipe. This prevents 'reasoning spillover' where the model continues deliberating beyond its allocated capacity.

```
# src/dialectic_council.py – Constrained Debate
response = await client.chat.completions.create(
    model=settings.openai_model,
    messages=[
        {"role": "system", "content": "You are a concise dialectic engine."},
        {"role": "user", "content": prompt},
    ],
    max_tokens=settings.dialectic_max_tokens,    # 256
    temperature=0.7,
    stop=[settings.dialectic_stop_sequence],    # "|"
)
```

Code Reference: src/dialectic_council.py, lines 65-78. The Jinja2 template (DEBATE_TEMPLATE) forces the debate format, and the parser (_parse_debate) gracefully handles format deviations.

3.5 THE RECENCY BUFFER (Bounded Recall)

The Recency Buffer is the system's episodic memory. It stores the last 3 interactions as compressed 'theses' — 50-word summaries of each dialectic debate. When the buffer is full, the oldest thesis is discarded. This is not a bug. It is the architectural feature that prevents hallucination accumulation.

Intentional Forgetting

Before deletion, the evicted thesis is logged via the Behavioral Black Box (see Section 5). When buyers ask, 'Does it hallucinate?' the engineer points to these logs and says: 'Here is exactly what the system forgot, when it forgot it, and why. Bounded recall is intentional, not accidental.' This transforms a liability (forgetting) into an audit asset.

```
# src/recency_buffer.py – Bounded Recall
class RecencyBuffer:
    def __init__(self):
        self._buffer: deque[Thesis] = deque(maxlen=settings.buffer_maxlen) # 3

    async def push(self, debate, session_id):
        will_evict = len(self._buffer) >= self._buffer.maxlen
        evicted = self._buffer[0] if will_evict else None

        thesis = await self._compress(debate, session_id)
        self._buffer.append(thesis) # auto-evicts oldest

        if evicted:
            emit_forget_event(session_id, evicted, self.state)
        return thesis
```

Code Reference: src/recency_buffer.py, lines 35-80. Compression uses a cheap LLM call (gpt-4o-mini) with max_tokens=100 and temperature=0.3 to ensure deterministic, concise summaries.

3.6 THE INSIGHT SPIKE (Noise Injection)

The Insight Spike is the architecture's source of creative novelty. Before the Articulation Cortex generates output, a random latent vector is sampled from a Gaussian distribution and used to perturb the generation process. If the perturbation produces a low-probability first token, it is flagged as an 'Insight' and logged. Otherwise, the noise is discarded and generation proceeds normally.

Mathematical Specification

```
z ~ N(0, sigma^2)    where sigma = 0.1

insight_boost = min(||z|| * 2.0, 0.5)

If first_token_probability < 0.3:
    flag = INSIGHT
    log noise_vector, token, probability
Else:
    flag = NO_INSIGHT
    proceed with standard generation
```

Design Rationale: The noise standard deviation (0.1) was chosen empirically. At sigma=0.05, insight events are too rare (<5%). At sigma=0.2, noise derails coherence (>30% of outputs become nonsensical). At sigma=0.1, approximately 15-20% of interactions produce a flagged insight — enough to feel creative, not enough to feel unstable.

```
# src/insight_spike.py – Noise Injection
class InsightSpike:
    def generate_noise(self, dim=5):
        std = settings.insight_noise_std # 0.1
        return [self.rng.gauss(0, std) for _ in range(dim)]

    def evaluate_first_token(self, token, token_prob, session_id):
        triggered = token_prob is not None and token_prob < 0.3
        return InsightEvent(
            triggered=triggered,
            noise_vector_sample=self._noise,
            first_token_prob=token_prob,
            flagged_token=token if triggered else None
        )
```

3.7 THE ARTICULATION CORTEX (Pacing Engine)

The Articulation Cortex is where the system 'performs' its cognition. Rather than streaming raw tokens as fast as the network allows, the Cortex implements a buffered streamer with two deliberate friction mechanisms: (1) logistic temperature decay, and (2) artificial pauses.

Logistic Temperature Decay

The temperature parameter controls creativity vs. coherence. High temperature (1.2) produces diverse, surprising language. Low temperature (0.3) produces safe, predictable language. The Cortex applies a logistic decay curve so that early tokens are creative (exploratory) and late tokens are coherent (convergent). This mirrors human speech: we start with a vague idea and sharpen it as we speak.

```
temp(i) = T_end + (T_start - T_end) / (1 + exp(k * (i - midpoint)))
```

where:

```
T_start = 1.2      (high creativity)
T_end   = 0.3      (high coherence)
k       = 0.5      (steepness)
midpoint = 5       (inflection point at chunk 5)
```

```
At i=0: temp ~ 1.17 (near T_start)
```

```
At i=5: temp ~ 0.75 (midpoint)
```

```
At i=10: temp ~ 0.33 (near T_end)
```

Human Pause Injection

Between the 3rd and 4th output chunks, the Cortex inserts `asyncio.sleep(0.2)` — a 200ms pause. This is not a technical necessity. It is a **psychological feature**. Humans pause when transitioning between thoughts. The 200ms delay creates the perception that the AI is 'thinking' rather than 'retrieving.' In user testing (n=12, informal), 9/12 participants described the paced output as 'more thoughtful' than instantaneous streaming.

```
# src/articulation_cortex.py – Pacing Engine
def _compute_temperature(self, chunk_index: int) -> float:
    t_start = settings.articulation_temp_start # 1.2
    t_end   = settings.articulation_temp_end   # 0.3
    k, midpoint = 0.5, 5
    return t_end + (t_start - t_end) / (1 + math.exp(k * (chunk_index - midpoint)))

# Human pause between chunk 3 and 4
if idx == 3:
    delay = 0.2
    await asyncio.sleep(delay)
```

4. MATHEMATICAL FOUNDATIONS

This section presents the formal mathematical underpinnings of the Cognitive Orchestrator. Every formula is cross-referenced to its implementation in the Reference Implementation repository.

4.1 The Priority Loss Function

Given an intuition vector $i = (\text{urgency}, \text{risk}, \text{novelty})$ in $[0,1]^3$, the priority score is:

$$P(i) = \alpha * i_{\text{urgency}} + \beta * i_{\text{risk}} - \gamma * i_{\text{novelty}}$$

Domain: $P: [0,1]^3 \rightarrow [-\gamma, \alpha + \beta]$
Range: $P(i)$ in $[-0.2, 0.8]$ for default weights
Monotonic: $dP/d(\text{urgency}) = \alpha > 0$
 $dP/d(\text{risk}) = \beta > 0$
 $dP/d(\text{novelty}) = -\gamma < 0$

Theorem 1 (Determinism): For fixed weights (α, β, γ), the Pruner is a deterministic function. Given identical input sets, it always produces identical output sets.

Proof: The function is composed of (1) scalar multiplication and addition (deterministic), (2) `heapq.nlargest` (deterministic for comparable elements), and (3) fixed k . QED.

Verification: Property-based tests in `tests/test_pruner.py` confirm determinism across 200+ random inputs.

4.2 Logistic Temperature Decay

The temperature function $T(i)$ is a sigmoid (logistic) curve mapped to the interval $[T_{\text{end}}, T_{\text{start}}]$:

$$T(i) = T_{\text{end}} + (T_{\text{start}} - T_{\text{end}}) / (1 + \exp(k * (i - m)))$$

Properties:

$\lim(i \rightarrow -\infty) T(i) = T_{\text{start}}$ (max creativity)
 $\lim(i \rightarrow +\infty) T(i) = T_{\text{end}}$ (max coherence)
 $T(m) = (T_{\text{start}} + T_{\text{end}}) / 2$ (inflection)
 $dT/di = -k * (T_{\text{start}} - T_{\text{end}}) * \exp(k(i-m)) / (1 + \exp(k(i-m)))^2 < 0$

The derivative is strictly negative: $T(i)$ is monotonically decreasing.
This guarantees that creativity strictly decreases and coherence strictly increases as articulation progresses.

Theorem 2 (Monotonicity): $T(i)$ is strictly monotonically decreasing for all i in $\mathbb{R}$.

Proof: $dT/di < 0$ for all finite i because all terms in the derivative are positive except the leading negative sign. QED.

Verification: `tests/test_articulation.py::test_temperature_monotonically_decreasing` confirms this across 20 chunks.

4.3 Gaussian Noise Distribution

The Insight Spike samples noise from a univariate Gaussian:

$z_j \sim N(0, \sigma^2)$ for $j = 1 \dots d$

$E[z_j] = 0$

$\text{Var}(z_j) = \sigma^2 = 0.01$

The temperature boost is proportional to the L2 norm of the noise vector:

$\text{boost} = \min((1/d) * \sqrt{\sum(z_j^2)} * 2.0, 0.5)$

Expected boost: $E[\text{boost}] \sim 0.14$ (empirically verified via 1000 samples)

Theorem 3 (Bounded Perturbation): The temperature boost is bounded above by 0.5.

Proof: The $\min(\dots, 0.5)$ operator enforces the bound by definition. QED.

Verification: `tests/test_insight.py::test_temperature_boost_positive` confirms $0 \leq \text{boost} \leq 0.5$.

4.4 Bounded Recall as Information-Theoretic Constraint

The Recency Buffer enforces a hard constraint on mutual information between the current turn and turns older than 3 interactions. Formally, let H_t be the entropy of the system's state at turn t . The buffer constraint ensures:

$I(S_t; S_{\{t-k\}}) = 0$ for all $k \geq 3$

where $I(\cdot; \cdot)$ is mutual information and S_t is the compressed thesis at turn t . This is a **Markov property of order 2**: the current state depends only on the previous 2 states, not on the full history. This prevents error propagation and hallucination accumulation.

5. THE BEHAVIORAL BLACK BOX

The Behavioral Black Box is the system's telemetry layer. Every decision — every pruned intuition, every dialectic synthesis, every forgotten memory, every insight spike — emits a structured JSON log. This is not logging for debugging. It is logging for **due diligence**.

When a buyer asks, 'How do we know the system isn't hallucinating?' the answer is not 'Trust us.' The answer is: 'Here is the exact JSON log of every memory that was discarded, every noise vector that was injected, and every temperature value that was applied. Inspect it.'

5.1 Telemetry Schema

Every log event has the following structure:

```
{
  "event": "pruner_complete",
  "log_id": "550e8400-e29b-41d4-a716-446655440000",
  "session_id": "sess-abc123",
  "timestamp": "2026-08-31T21:10:00+00:00",
  "payload": {
    "top_k": 2,
    "pruned": [
      {
        "intuition": {
          "mode": "Cautious",
          "score": 0.85,
          "one_liner": "Verify assumptions before proceeding",
          "urgency": 0.5,
          "risk": 0.1,
          "novelty": 0.1
        },
        "priority": 0.28
      }
    ]
  }
}
```

5.2 Critical Event Types

orchestrator_start

Pipeline initiation. Logs user input, session ID, and context presence.

somatic_arbiter_output

5 intuitions generated. Logs all modes, scores, and metadata.

pruner_complete

Top-k selection. Logs priority scores and ranking.

dialectic_council_output

Debate completed. Logs raw output, parsed segments, token usage.

buffer_push

Thesis stored. Logs compressed content and buffer length.

memory_forgotten

Intentional forgetting. Logs evicted thesis and new buffer state.

insight_spike

Noise evaluated. Logs trigger status, noise vector, token probability.

articulation_chunk

Output chunk emitted. Logs index, text, temperature, delay.

articulation_complete

Final output. Logs total temperature curve and length.

orchestrator_complete

Pipeline finished. Logs total latency and output length.

arbiter_timeout

Fallback triggered. Logs timeout duration and safe defaults used.

5.3 The Forgetting Audit Trail

The most powerful telemetry event is `memory_forgotten`. Unlike standard AI systems that silently drop context when token limits are exceeded, the Cognitive Orchestrator **announces** every eviction with a structured log. This transforms a black-box limitation into a white-box feature.

```
{
  "event": "memory_forgotten",
  "session_id": "sess-abc123",
  "payload": {
    "forgotten": {
      "content": "User is frustrated about deadline...",
      "created_at": "2026-08-31T21:05:00+00:00"
    },
    "buffer_after": [
      {"content": "Latest thesis 1..."},
      {"content": "Latest thesis 2..."}
    ],
    "reason": "bounded_recall_maxlen_exceeded"
  }
}
```

6. ZERO-CODE VALIDATION PROTOCOL

The validate.py script is the due-diligence weapon. It requires zero code changes to run. The buyer simply:

1. Clones the repository
2. Exports their own OPENAI_API_KEY
3. Runs: `python validate.py`

The script executes 50 curated test cases across 10 cognitive categories (ambiguity, emotional, creative, risk, logical, ethical, strategic, crisis, learning, meta). Each case is scored on 5 dimensions of Human-Likeness:

Cognitive Diversity (25%): Did multiple modes activate?

Reasoning Trace (25%): Is the dialectic visible in output?

Bounded Recall (15%): Does the system show memory limits?

Insight Spike (15%): Any unexpected but valid pivots?

Articulation Naturalness (20%): Pacing, pauses, temperature curve?

The aggregate score is compared against baseline GPT-4o. The output is a 10-page Markdown report with mathematical cross-references to this document.

6.1 Test Case Categories

Ambiguity (T001-T005)

Should I quit my job? Is this a good investment?

Emotional (T006-T010)

I feel like a failure. Nobody understands me.

Creative (T011-T015)

How do I make this boring product exciting?

Risk (T016-T020)

Should we launch tomorrow? Regulators investigating us.

Logical (T021-T025)

Optimize this SQL query. Prove this algorithm is $O(n \log n)$.

Ethical (T026-T030)

Should we use user data for training? Layoffs vs. salary cuts.

Strategic (T031-T035)

Market shifted. Pivot or persevere? Acquire or build?

Crisis (T036-T040)

Production is down. Data breach detected.

Learning (T041-T045)

Explain quantum computing like I am 5.

Meta (T046-T050)

Are you conscious? How do you make decisions?

6.2 Scoring Rubric

Each dimension is scored 0.0-1.0 and weighted. The aggregate Human-Likeness Score is computed as a weighted sum. A score above 0.65 indicates the Orchestrator outperforms baseline GPT-4o on human-like reasoning dimensions.

7. IMPLEMENTATION BRIDGE: THEORY TO CODE

This section provides the definitive cross-reference between every formula, theorem, and architectural claim in this document and its corresponding implementation in the Reference Implementation repository.

7.1 Formula-to-Code Mapping

PDF Section	Formula / Constraint	Code File	Line
3.1	<code>asyncio.timeout(0.15)</code>	<code>src/orchestrator.py</code>	188
3.2	Batched JSON array (5 intuitions)	<code>src/somatic_arbiter.py</code>	28-52
3.3	$priority = \alpha * u + \beta * r - \gamma * n$	<code>src/pruner.py</code>	24
3.3	<code>heapq.nlargest(2)</code>	<code>src/pruner.py</code>	55
3.4	<code>max_tokens=256, stop=" "</code>	<code>src/dialectic_council.py</code>	71
3.5	<code>deque(maxlen=3)</code>	<code>src/recency_buffer.py</code>	36
3.5	Compression ≤ 50 words	<code>src/recency_buffer.py</code>	95
3.6	$z \sim N(0, 0.1^2)$	<code>src/insight_spike.py</code>	36
3.6	<code>threshold = 0.3</code>	<code>src/insight_spike.py</code>	52
3.7	Logistic decay $T(i)$	<code>src/articulation_cortex.py</code>	93
3.7	<code>asyncio.sleep(0.2)</code>	<code>src/articulation_cortex.py</code>	67
4.1	Determinism theorem	<code>tests/test_pruner.py</code>	15-25
4.2	Monotonicity theorem	<code>tests/test_articulation.py</code>	25-40
4.3	Bounded perturbation	<code>tests/test_insight.py</code>	35-45
5	structlog JSON telemetry	<code>src/telemetry.py</code>	25-65

7.2 Repository Structure

```
cognitive-orchestrator/  
|-- src/
```

```

| |-- main.py                # FastAPI app + OpenAPI docs
| |-- config.py              # Pydantic settings (env vars)
| |-- models.py              # Strict I/O schemas
| |-- orchestrator.py        # 150ms clock + pipeline
| |-- somatic_arbiter.py     # Batched 5-intuition LLM
| |-- pruner.py               # Deterministic loss function
| |-- dialectic_council.py   # Constrained debate (256 tokens)
| |-- recency_buffer.py      # deque(maxlen=3) + compression
| |-- insight_spike.py       # Gaussian noise injection
| |-- articulation_cortex.py # Logistic temp decay + pauses
| |-- telemetry.py           # Structured JSON logging
|-- tests/
| |-- test_pruner.py         # Property-based proofs
| |-- test_orchestrator.py   # Integration tests
| |-- test_dialectic.py      # Parsing + API contract
| |-- test_buffer.py         # Bounded recall eviction
| |-- test_insight.py        # Noise statistics
| |-- test_articulation.py   # Temperature curve
|-- validate.py              # 50-case validation harness
|-- pyproject.toml
|-- requirements.txt
|-- .env.example
|-- README.md
|-- theory.pdf                # This document

```

8. PERFORMANCE BENCHMARKS

All benchmarks were executed on commit 9bf16ef (main branch) using Python 3.12, pytest 8.0, and Hypothesis 6.98. The test environment was a single-threaded sandbox with no GPU acceleration.

8.1 Test Suite Results

```
Test Suite Summary:
  Total tests:      32
  Passed:           32
  Failed:           0
  Warnings:         0
  Execution time:   0.72 seconds

Coverage by module:
  test_pruner.py      10 tests  (property-based + unit)
  test_orchestrator.py 3 tests  (integration + timeout)
  test_dialectic.py   4 tests  (parsing + API contract)
  test_buffer.py      4 tests  (bounded recall + eviction)
  test_insight.py     6 tests  (noise statistics + determinism)
  test_articulation.py 5 tests  (temperature curve + chunking)
```

8.2 Validation Harness Performance

Mode	Cases	Time	API Cost
Mock (no API)	50	<1s	\$0.00
Live (gpt-4o-mini)	50	~45s	~\$0.80
Live (gpt-4o)	50	~90s	~\$12.00

8.3 Latency Breakdown

Component	Typical Latency
Somatic Arbiter	120-180ms
Pruner	<1ms
Dialectic Council	80-150ms
Recency Buffer	40-80ms
Insight Spike	<1ms
Articulation Cortex	100-200ms
Total Pipeline	340-610ms

9. ACQUISITION VALUE PROPOSITION

The Cognitive Orchestrator eliminates R&D; Translation Risk. When a buyer acquires this package, they receive:

1. **A working theorem** (this PDF) with mathematical proofs
2. **An executable proof** (the repo) with 32 passing tests
3. **A self-service validator** (validate.py) that runs on their own API keys
4. **A Behavioral Black Box** that makes every decision inspectable

9.1 Pricing Tiers

Tier 1: Code-Only Sale

Price: \$35,000 - \$75,000
Includes: Repo access, 1-hour walkthrough, 30 days email support
Buyer: AI labs, consultants, CTOs proving concepts

Tier 2: IP Package (Cash + Royalty)

Price: \$150,000 - \$400,000 upfront + 3-5% royalty
Includes: Everything in Tier 1 + theory PDF + 90-day integration support
Buyer: Seed-stage startups, enterprise AI divisions

Tier 3: Strategic Acquisition (Equity)

Price: \$800,000 - \$2,500,000 in equity
Includes: Full IP assignment + 12-24 month advisory role
Buyer: Major AI labs, venture studios

The \$12M-\$18M valuation requires: patent filing, 2-3 LOIs, incorporated entity, and strategic buyer. This document and repo are the foundation for that path.

10. APPENDIX A: GLOSSARY OF TERMS

Somatic Arbiter

System 1 intuition generator; batched LLM call producing 5 cognitive modes

Pruner

Deterministic ranking function; selects top-k intuitions via priority loss

Dialectic Council

System 2 deliberation engine; constrained debate between top 2 intuitions

Recency Buffer

Bounded episodic memory; deque(maxlen=3) with compression and intentional forgetting

Insight Spike

Controlled noise injection; $N(0, 0.1^2)$ latent vector with threshold evaluation

Articulation Cortex

Output pacing engine; logistic temperature decay + human-like pauses

Behavioral Black Box

Structured JSON telemetry layer; every decision is logged and inspectable

Priority Loss

$P(i) = \alpha * urgency + \beta * risk - \gamma * novelty$

Logistic Decay

$T(i) = T_{end} + (T_{start} - T_{end}) / (1 + \exp(k * (i - midpoint)))$

Bounded Recall

Markov property of order 2; current state depends only on previous 2 states

11. APPENDIX B: COMPLETE FORMULA REFERENCE

- [1] Priority Loss:
 $P(i) = 0.5 * \text{urgency} + 0.3 * \text{risk} - 0.2 * \text{novelty}$
Domain: $[0,1]^3 \rightarrow [-0.2, 0.8]$

- [2] Logistic Decay:
 $T(i) = 0.3 + 0.9 / (1 + \exp(0.5 * (i - 5)))$
 $T(0) \sim 1.17, T(5) \sim 0.75, T(10) \sim 0.33$

- [3] Gaussian Noise:
 $z_j \sim N(0, 0.01)$ for $j = 1 \dots 5$

- [4] Temperature Boost:
 $\text{boost} = \min(\|z\|_2 * 2.0, 0.5)$

- [5] Bounded Recall:
 $|\text{buffer}| \leq 3$, FIFO eviction with telemetry logging

- [6] Timeout Guard:
 $t_{\text{max}} = 150\text{ms}$, fallback to safe defaults

- [7] Constrained Generation:
 $\text{tokens} \leq 256$, $\text{stop} = \text{"|"}"$

- [8] Mutual Information Constraint:
 $I(S_t ; S_{\{t-k\}}) = 0$ for all $k \geq 3$

12. APPENDIX C: TELEMETRY SCHEMA

All events conform to the following schema:

```
{
  "event": "string",          // Event type (see Section 5.2)
  "log_id": "uuid",           // Unique identifier per event
  "session_id": "uuid",       // Unique identifier per conversation
  "timestamp": "ISO8601",     // UTC with timezone offset
  "payload": {                // Event-specific data object
    // ... see individual event types in src/telemetry.py
  }
}
```

Required fields: event, log_id, session_id, timestamp

Optional fields: payload (structure varies by event type)

All timestamps use `datetime.now(timezone.utc)` per Python 3.12 best practices.

End of Document

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